Functional Equations — problem 11.34

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The function f is defined on the set of positive integers as follows :

$$f(1)=1,\quad f(3)=3,\quad f(2n)=f(n),\quad f(4n+1)=2f(2n+1)-f(n),\quad f(4n+3)=3f(2n+1)-2f(n).$$

Find all values of n with $f(n)=n$ and $1\leq n\leq 1988$ (IMO 1988).